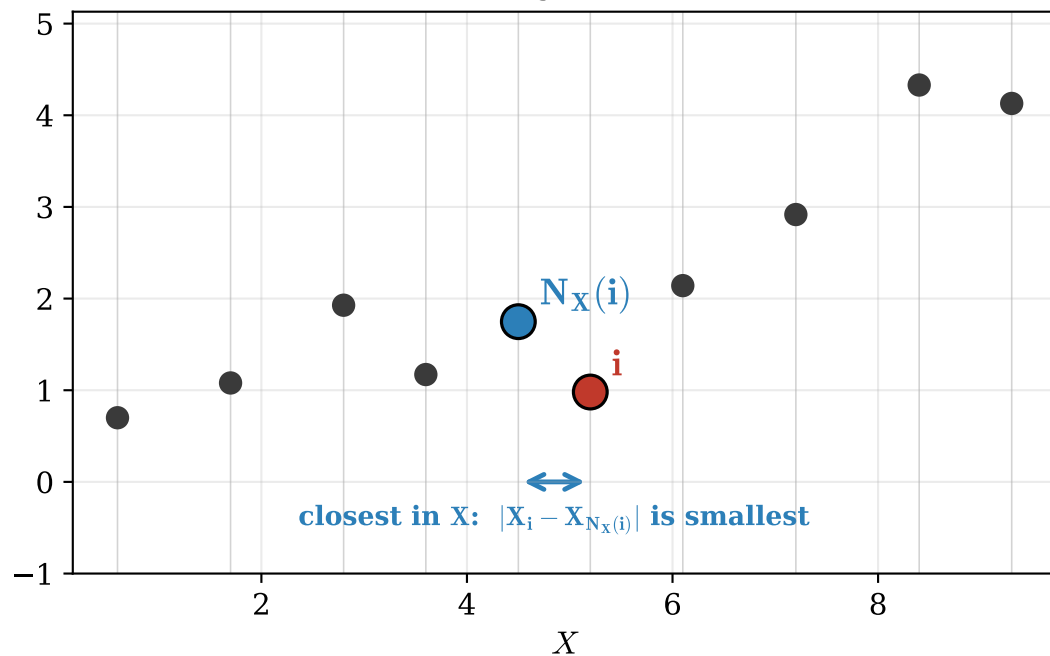


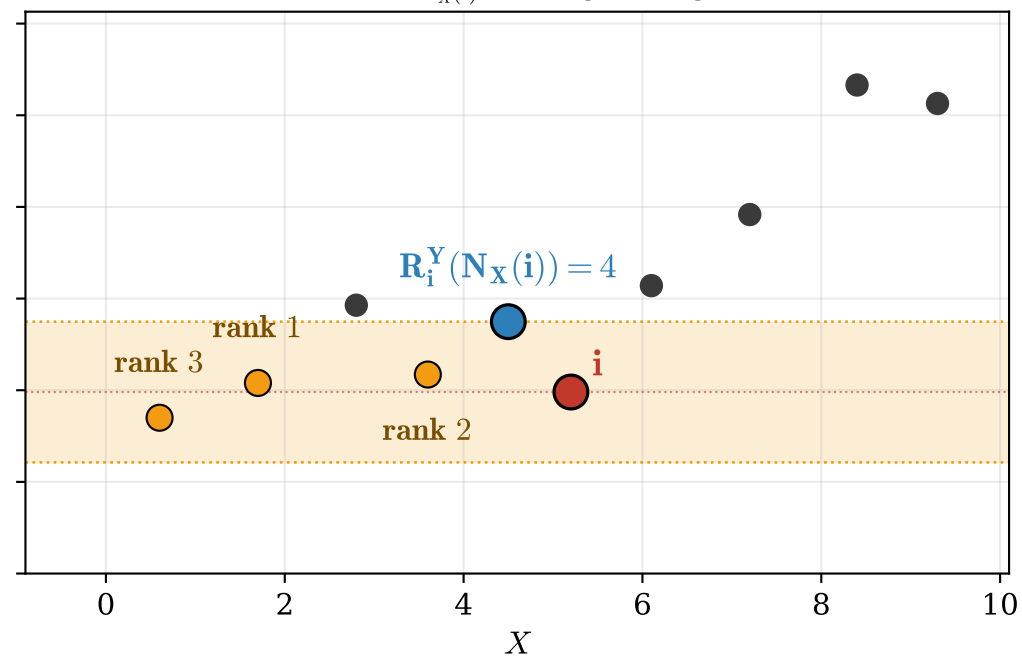
Information Imbalance:  $\widehat{\Pi}_n(X \rightarrow Y) = \frac{2}{n^2} \sum_{i=1}^n R_i^Y(N_X(i))$

Panel A: nearest neighbour of  $X_i$  in  $X$ -space



●  $(X_i, Y_i)$ 
●  $(X_{N_X(i)}, Y_{N_X(i)})$

Panel B: rank of  $Y_{N_X(i)}$  among  $Y$ -neighbours of  $Y_i$



● inside band (3 pts)
 ●  $(X_i, Y_i)$ 
●  $(X_{N_X(i)}, Y_{N_X(i)})$